

1 Lecture 10b: Interactivity and a quick introduction to Dash library

Data Visualization · 1-DAV-105

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```
[ ]: # importing libraries
import numpy as np
import pandas as pd
import plotly.express as px
```

1.1 Interactivity

Interactive visualization engages audience, allows them to explore data in depth and according to their interest.

1.1.1 Examples

- US cities with the same name ([website](#)), see also the animated explainer
- PhD gender gap ([website](#))
- Making it big ([website](#)), more animated than interactive

1.1.2 Techniques in interactivity visualization

Similar to decisions made in designing a static plot:

- Selecting variables (x, y, color, ...)
- Filtering data (selecting table rows)
- Highlighting points or groups
- Aggregating (display countries or region summaries)
- Zooming / panning
- Rescaling (log-scale) / reexpressing (e.g. % instead of counts)
- Sorting (e.g. bars in bargraphs)
- Displaying details (tooltips)
- Annotating
- Bookmarking

(Stephen Few)

1.1.3 Dashboard

- A display consisting of multiple plots, summarizing the current state of important indicators (e.g. of a business, pandemics, ...)
- Inspired by dashboards in cars and planes
- Often interactive, but main features in default view

Two SARS-CoV-2 examples:

- [WHO](#)
- [Nextstrain](#)

- many options: selecting color, filtering, highlighting, aggregating, zooming and panning (maps and tree), rescaling (time vs divergence), tooltips, bookmarking

1.1.4 Interactivity in Plotly Express

All Plotly plots by default have some interactivity:

- Filtering groups
- Zooming / panning
- [Details](#)
- [Spike lines](#)

Example 1: switching off categories Country indicators from World Bank, <https://databank.worldbank.org/home> under CC BY 4.0 license. Regions can be switched on and off.

```
[ ]: url = 'https://fmfi-compbio.github.io/viz/data/World_bank.csv'
countries = pd.read_csv(url)

px.scatter(
    countries, x="GDP2020", y="Expectancy2020", color="Region",
    hover_data=['Country'],
    title="Country indicators 2020", log_x=True,
    width=800, height=500
)
```

1.1.5 Example 2: Spike lines

Life expectancy data, based on free data from World bank via gapminder.org, CC-BY license (years 1900-2017) and [World bank](#) directly (years 2018-2021).

Compare data along a chosen year.

```
[ ]: url = "https://fmfi-compbio.github.io/viz/data/life_expectancy_years.csv"
orig_expectancy = pd.read_csv(url, index_col=0)
orig_expectancy = orig_expectancy.iloc[:, 1:].reset_index()
expectancy = pd.melt(orig_expectancy, id_vars=["Country"], var_name="Year",
    value_name="Expectancy")
expectancy['Year'] = expectancy['Year'].astype(int)
display(expectancy.head())

[ ]: selected = expectancy.query("Country=='Slovak Republic' or Country=='Portugal'")
fig=px.line(
    selected, x="Year", y="Expectancy", color="Country",
    width=800, height=500
)
fig.update_layout(hovermode="x unified")
```

1.1.6 Next: More interaction with Dash by Plotly

- Dash library by Plotly allows adding control elements (selectors, sliders, buttons, ...)

1.2 Data for Dash: a simple table of function values

Function `np.linspace` below creates a list of 5 evenly spaced numbers in interval $[1, 3]$, stored as an object of `array` class from the Numpy library.

```
[ ]: x = np.linspace(start=1, stop=3, num=5)
      print('x:', x)
```

The function below gets values of `x` and a dictionary of named mathematical functions, each consisting of function values of each value of `x` and converts this into a long `DataFrame` with columns `x`, `value`, and `function`. Zeroes and negative values are masked as missing to avoid problems with logarithmic `y` axis.

```
[ ]: def convert_table(x, function_dict):
      """ x is a list (or Numpy array) of values of x,
          function_dict is a dictionary containing function names as keys
          and lists of function values as values. The result will be a Pandas
          DataFrame (table) with each row containing triple x, function, value.
          Zeroes and negative values are masked as missing
          to avoid problems with logarithmic y axis. """

      # check that all functions have the same number of values as x
      for f in function_dict:
          assert(len(function_dict[f])==len(x))

      # create a wide table with each function as one column
      functions_wide = pd.DataFrame(function_dict, index=x)
      # reformat to long format
      # where each row is a triple x, function name, function value
      functions = (functions_wide.reset_index()
                   .melt(id_vars='index')
                   .rename(columns={'variable':'function', 'index':'x'}))

      # mask values <= 0 as missing values
      val = functions['value']
      functions['value'] = val.mask(val <= 0, np.nan)
      return functions
```

```
[ ]: functions = convert_table(x, {'quadratic': x * x, 'cubic': x * x * x})
```

Let us look at the resulting table functions:

- It has three columns named `'x'`, `'function'` and `'value'`.
- Each row is a triple, containing a function name and the values of x and $f(x)$.
- E.g. one of the rows for the cubic function has $x = 2$ and $f(x) = 2^3 = 8$.

```
[ ]: display(functions)
```

This DataFrame can be easily plotted using the Plotly library.

```
[ ]: figure = px.line(functions, x="x", y="value", color='function')
figure.show()
```

1.3 Interactive plots in Plotly Dash

- Dash library by Plotly allows adding control elements (selectors, sliders, buttons, ...).
- It is not preinstalled in Colab, so the next line will install it.

```
[ ]: # if needed, uncomment this command to install the required version of Dash
↳ library
# ! pip install dash==3.2.0
```

- The code below creates an interactive plot in which the user can choose which functions from the list to display.
- The code has many comments so read through it carefully.

```
[ ]: from dash import Dash, dcc, html, Input, Output

# create a list of all functions
all_functions = list(functions['function'].unique())

# create a new dash application app
app = Dash(__name__)

# Create layout of items in application
# one html <div> item containing text as small headers (H4),
# items for individual inputs and a graph at the bottom
# Currently we have two inputs:
# an input field for entering title text
# checkboxes for selecting functions
# These elements have identifiers which will be used later in the code
app.layout = html.Div([
    html.H4("Plot title: "),
    # input field for entering title text:
    dcc.Input(
        id='graph-title',
        type='text',
        value='My plot' # initial value
    ),
    html.H4("Select functions: "),
    # checkboxes for selecting functions:
    dcc.Checklist(
        id='selected-functions',
        options=all_functions,
        value=['quadratic'], # initial value, quadratic function is selected
        inline=True # place checkboxes horizontally
    )
])
```

```

    ),
    # graph itself
    dcc.Graph(id='graph-content')
])

# @app.callback is a function decorator applied to function update_figure below.
# It defines that this function will be called to update the graph
#     when the user makes a change.
#     Input will be the value entered to the input field with id graph-title and
#     the list of functions selected in dcc.Checklist object with id
#     ↪ 'selected-functions'.
#     Output will be the graph created by the function update_figure below,
#     which will be used to update dcc.Graph object with id 'graph-content'
@app.callback(
    Output('graph-content', 'figure'),
    [Input('graph-title', 'value'),
     Input('selected-functions', 'value')]
)
def update_figure(title, selected_functions):
    """ Function for plotting the functions listed in list selected_functions
    with plot title given in title"""

    # select a subset of functions table with just those functions in input list
    functions_subset = functions.query('function in @selected_functions')

    # create a plotly line plot using the smaller table in functions_subset
    figure = px.line(
        functions_subset,
        x="x", y="value", color="function",
        width=800, height=500
    )

    # add title to the plot
    figure.update_layout(title_text=title)

    return figure

# run the whole application
app.run_server(mode='inline')
pass

```